Ricky Tjandra, PhD

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Summary

Medical device product leader with a PhD in nanotechnology and end-to-end ownership across device development, regulatory strategy and commercial execution. Took a biosensor platform from concept to FDA 510(k) clearance (K243965), built multi-market regulatory strategy across FDA, Health Canada and GCC jurisdictions, and leads a cross-functional team spanning hardware, software, data science, clinical operations and quality.

Experience

FluidAI Medical (formerly NERv Technology) · Waterloo, ON

Vice President, Medical Devices

SEP 2019–PRESENT

- **Product roadmap & NPI:** Led development and commercialization of the Origin biosensor platform from concept and feasibility through clinical validation to FDA 510(k) clearance (K243965). Maintain a prioritized roadmap across an IVD sensor platform, a clinical decision support algorithm (ALRC) and a patient monitoring device. Lifecycle strategy including COGS reduction and claims evolution aligned to cleared indications.
- **Business strategy:** Built quantitative business cases and long-range financial projections supporting Series B fundraising. Wrote US marketing guidelines after 510(k) clearance defining permissible claims. Cultivated surgeon KOL, hospital and academic partnerships.
- **Leadership:** Player/coach lead across hardware, firmware, software, data science, clinical operations and quality; manage and coach product and QA associates. Present strategy and evidence to the board, investors and regulators.
- **Regulatory & quality:** Own regulatory filings and agency engagement across US FDA, Health Canada, UAE MOHAP and Saudi FDA, managing submissions and external consultants. Key member of the team that set up and maintains the ISO 13485:2016-aligned QMS.
- **R&D investment & IP:** Hold the Medical Device business unit budget. Secured competitive external funding (Mitacs, NSERC Alliance, Digital Supercluster); oversee SR&ED classification. Grew the patent portfolio from 3 to 18 active families.

Education

University of Waterloo

- **PhD, Chemical Engineering – Nanotechnology** 2015–2019
  - Waterloo Institute for Nanotechnology; 20 peer-reviewed publications, 955+ citations.
- **MASc, Chemical Engineering – Nanotechnology** 2014–2015
- **BASc, Nanotechnology Engineering** 2009–2014

Key skills

REGULATORY

FDA 510(k), De Novo, Q-Sub, Health Canada, SFDA, UAE MOHAP, SaMD / CDS

QUALITY

ISO 13485:2016, 21 CFR 820 (QMSR), 21 CFR Part 11, ISO 14971, design controls

PRODUCT & IP

Product strategy, PRDs/SRSs, roadmaps, business cases, patent strategy, FTO, technical due diligence

TECHNICAL

Biosensors, electrochemical sensing, data analysis (Python), Agile, Jira, Confluence